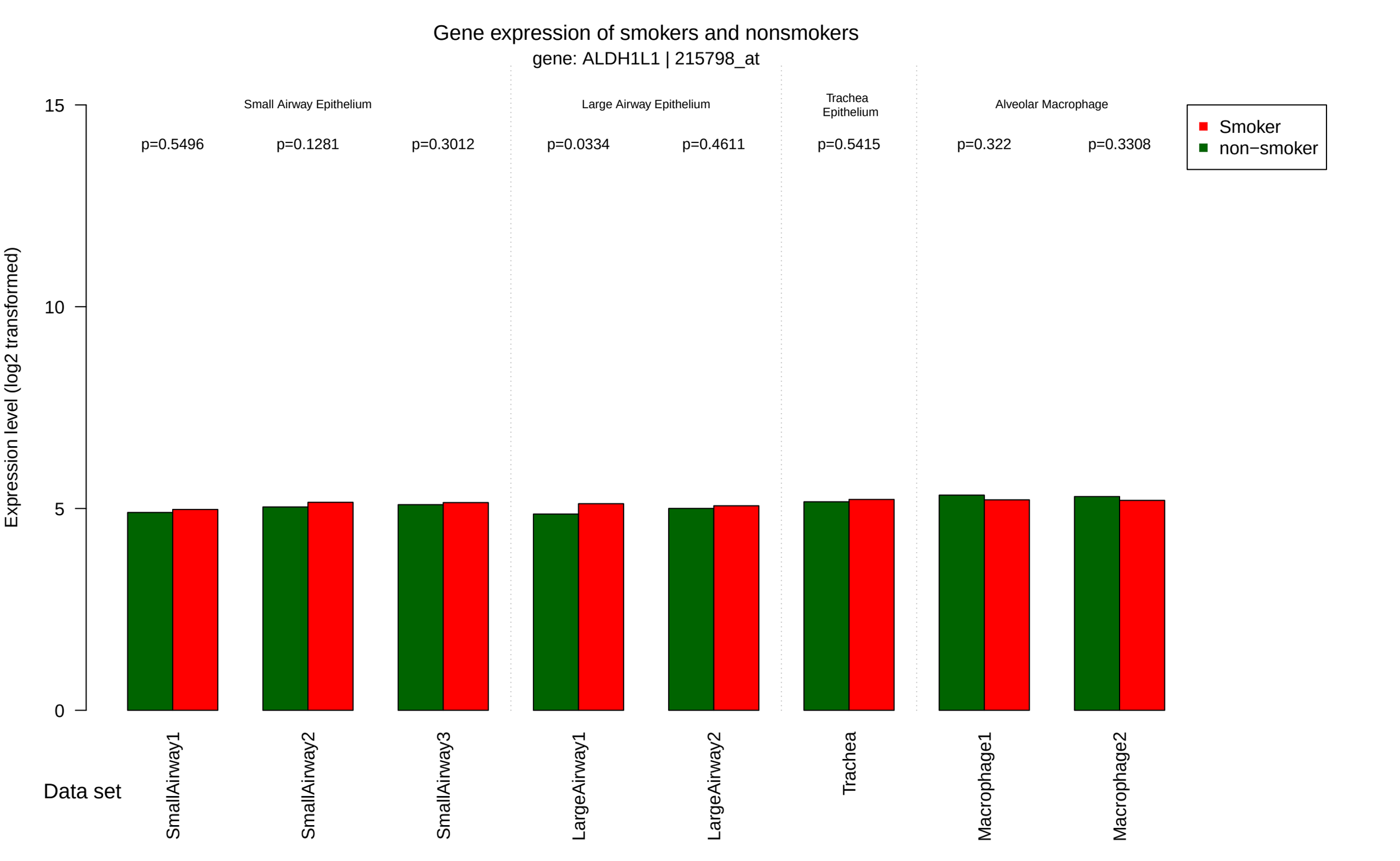
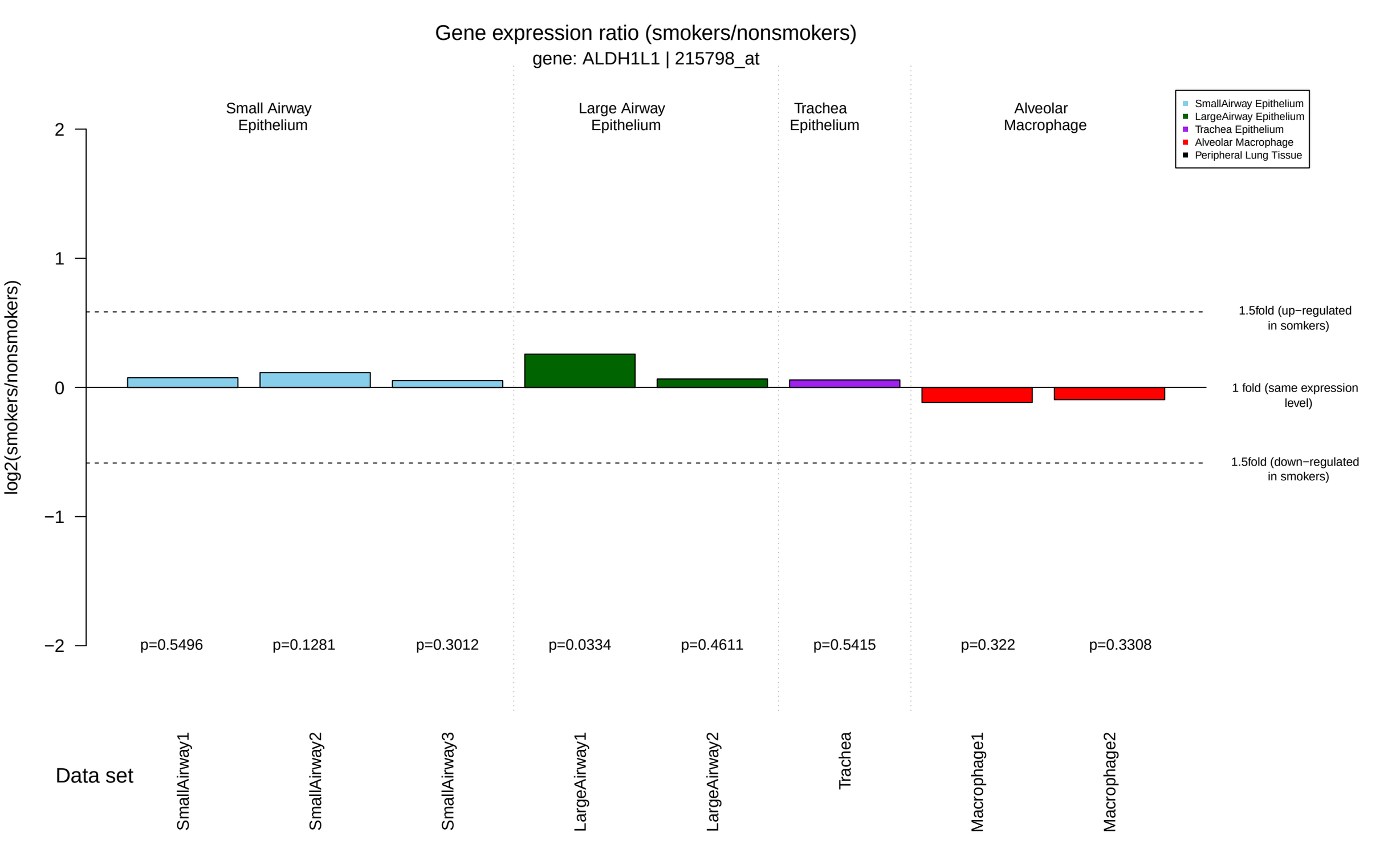
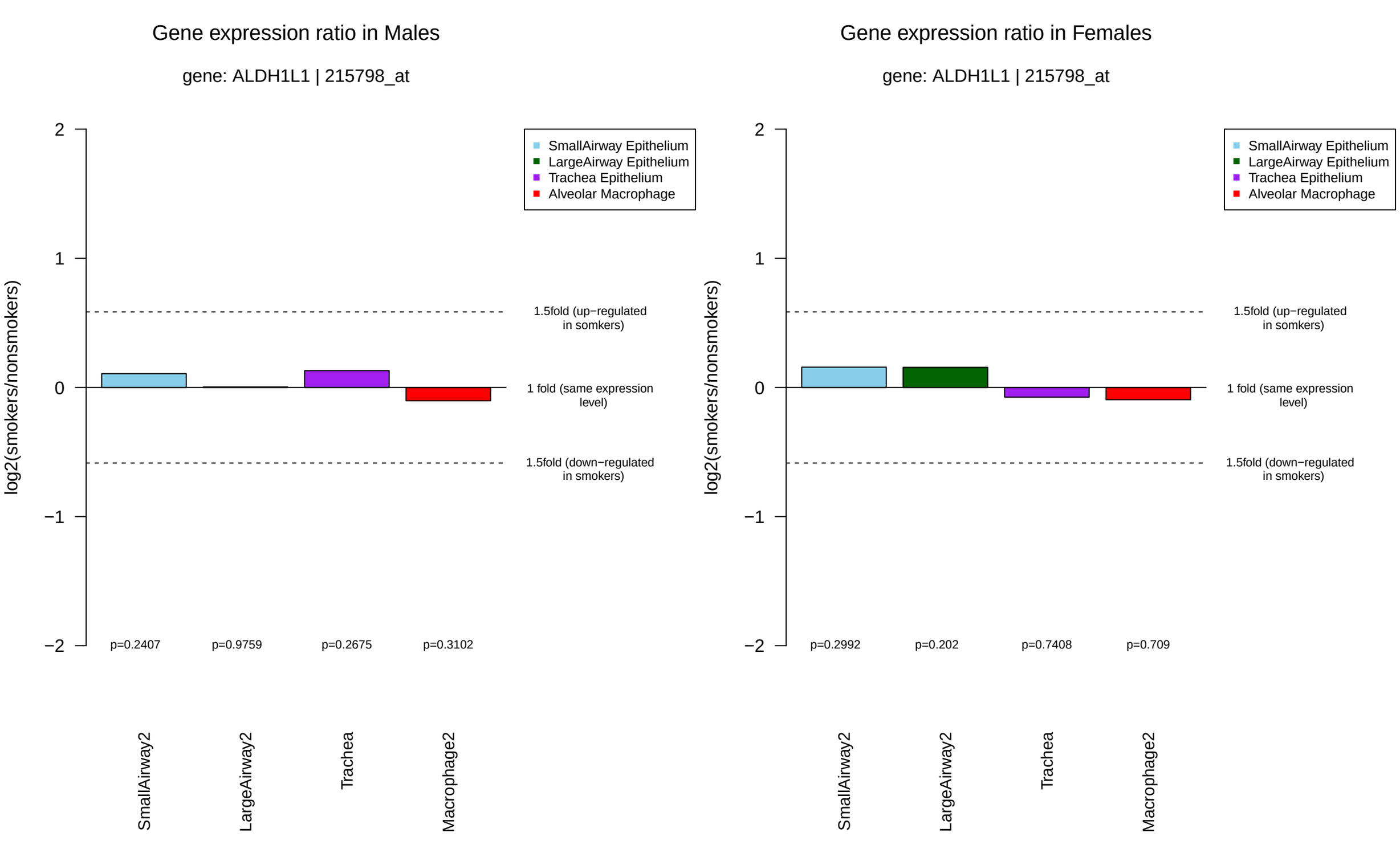
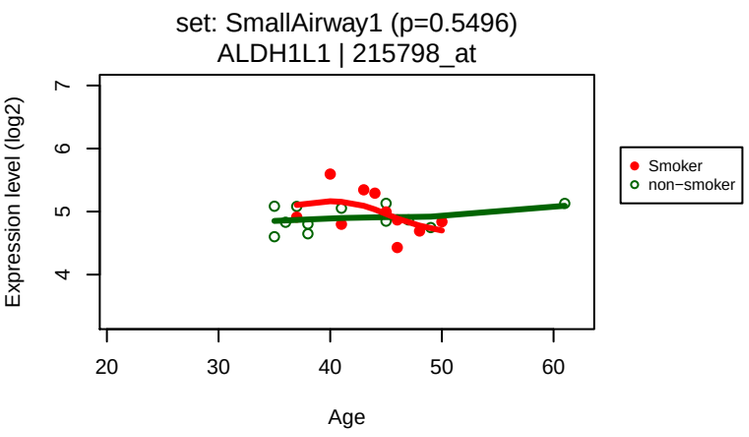
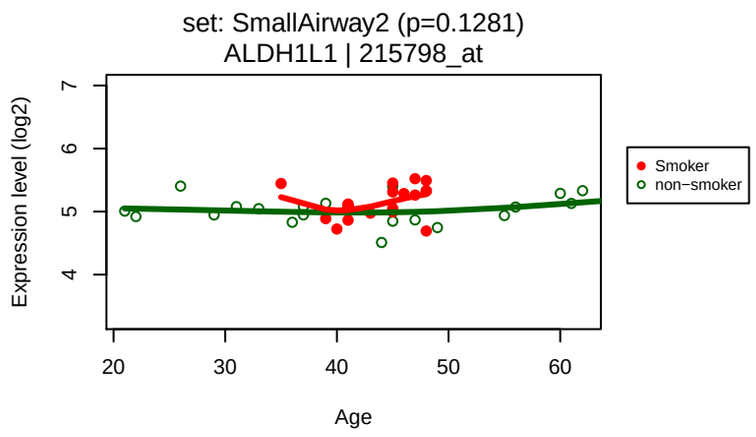
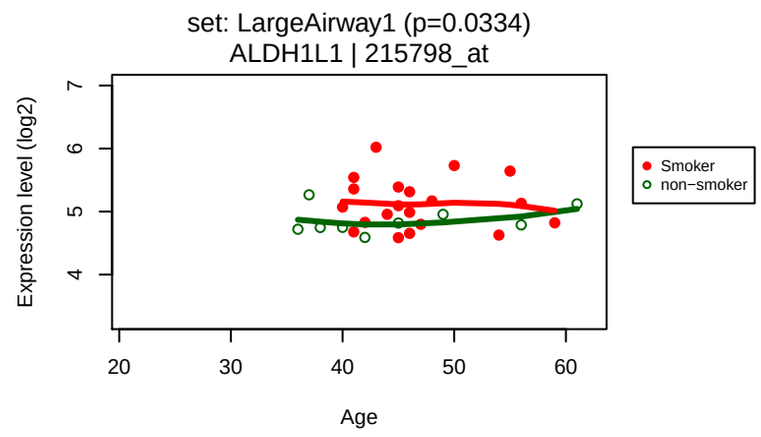

Gene expression of smokers and nonsmokers
gene: ALDH1L1 | 215798_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5496
p=0.1281
p=0.3012
p=0.0334
p=0.4611
p=0.5415
p=0.322
p=0.3308
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

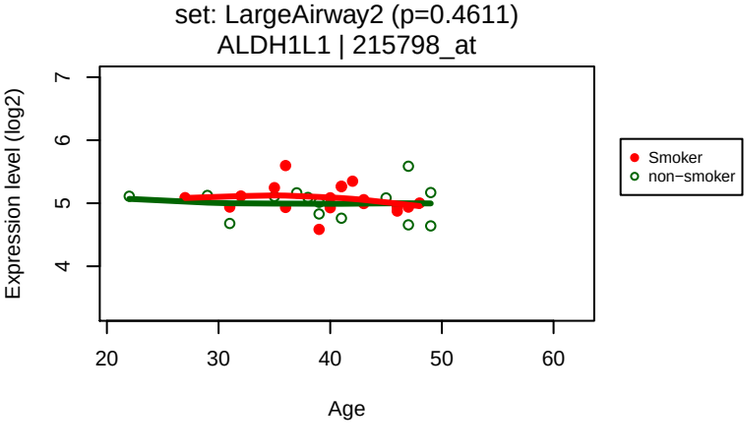

Gene expression ratio (smokers/nonsmokers)
gene: ALDH1L1 | 215798_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5496
p=0.1281
p=0.3012
p=0.0334
p=0.4611
p=0.5415
p=0.322
p=0.3308
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

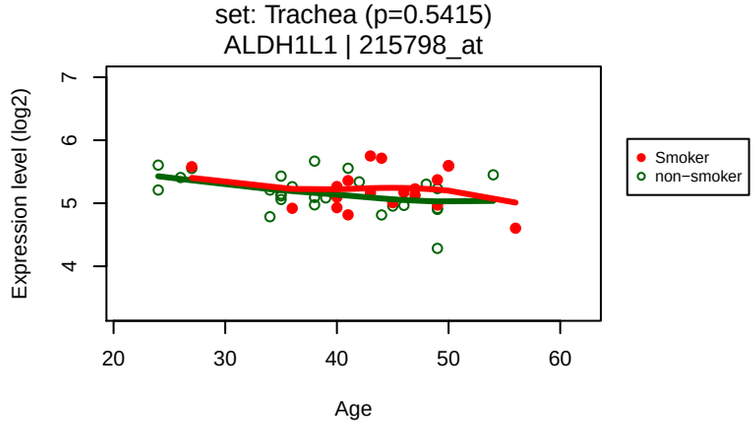

Gene expression ratio in Males
gene: ALDH1L1 | 215798_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2407
p=0.9759
p=0.2675
p=0.3102
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH1L1 | 215798_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2992
p=0.202
p=0.7408
p=0.709
SmallAirway2
LargeAirway2
Trachea
Macrophage2

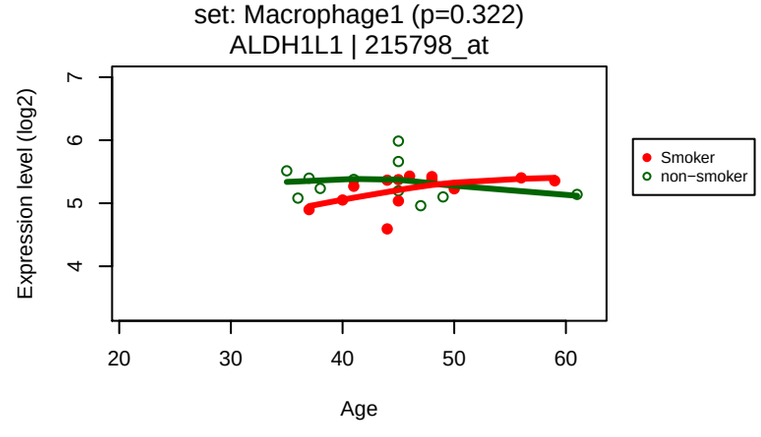

set: SmallAirway1 (p=0.5496)
ALDH1L1 | 215798_at
Expression level (log2)
Age
Smoker
non−smoker

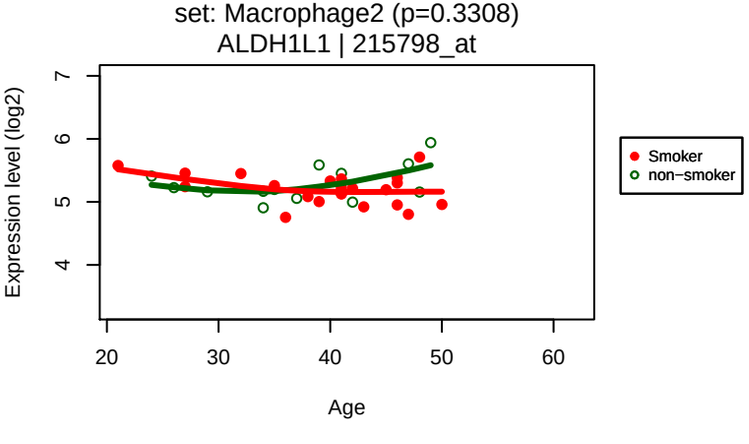

set: SmallAirway2 (p=0.1281)
ALDH1L1 | 215798_at
Expression level (log2)
Age
Smoker
non−smoker

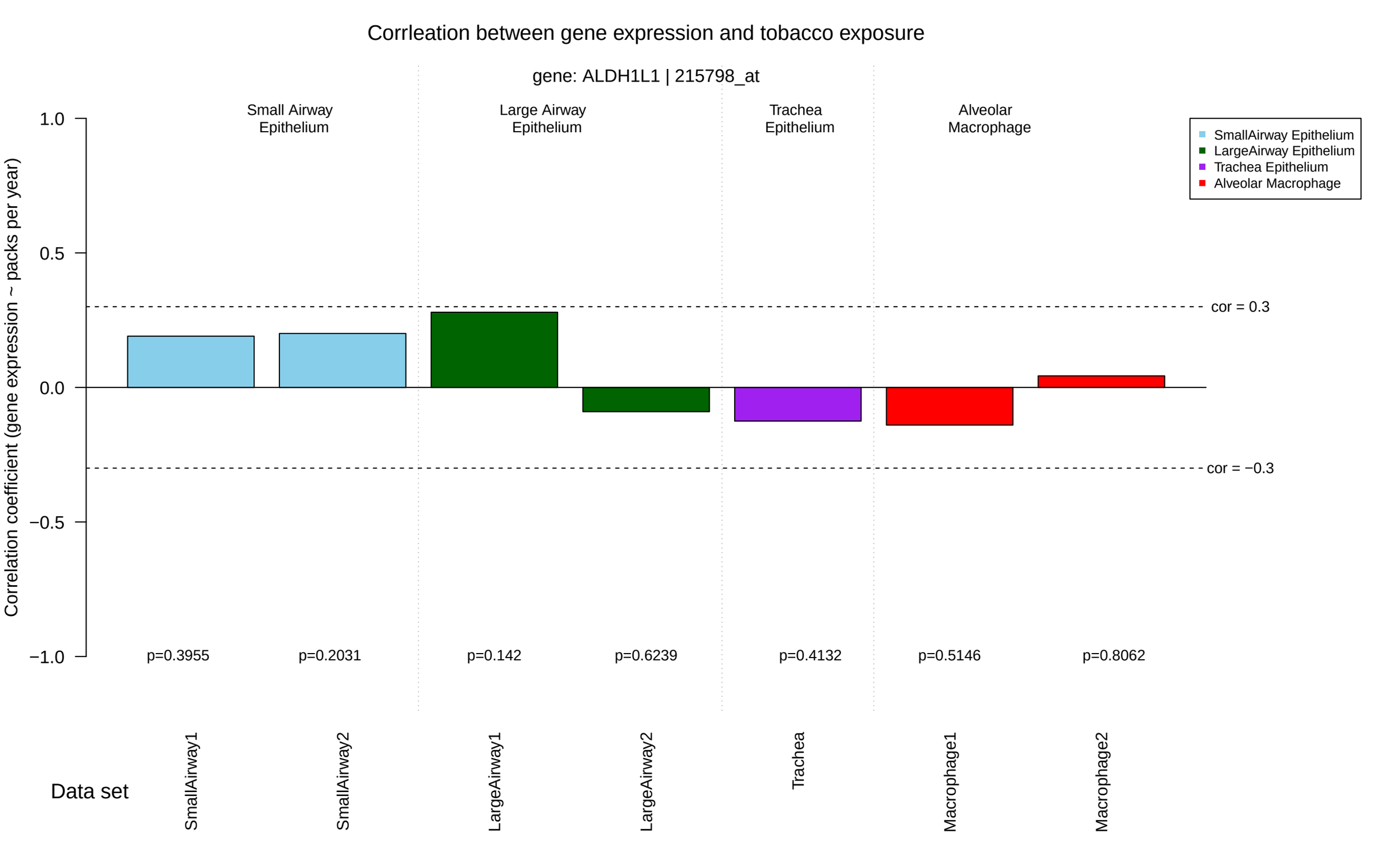

set: LargeAirway1 (p=0.0334)
ALDH1L1 | 215798_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4611)
ALDH1L1 | 215798_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5415)
ALDH1L1 | 215798_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.322)
ALDH1L1 | 215798_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3308)
ALDH1L1 | 215798_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH1L1 | 215798_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3955
p=0.2031
p=0.142
p=0.6239
p=0.4132
p=0.5146
p=0.8062
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2